CS Alumni Dashboard- Project Proposal

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Project Overview

Purdue Computer Science students often lack clear, accessible information on what career paths are possible with their degree. Existing data on alumni outcomes is outdated, scattered, or generalized, making it difficult for students to plan their academic tracks with the help of real-world data. Advisors, too, lack a centralized tool to guide students using concrete, data-backed trends. Additionally, by addressing this problem, the department has an opportunity to strengthen relationships with corporate partners by highlighting the success and placement of its graduates.

We propose a unified, interactive data platform designed for Purdue's Department of Computer Science in collaboration with the CCO. The platform will serve two key user groups: (1) students, advisors, and visitors looking to explore alumni career outcomes and trends, and (2) administrators managing data inputs, generating reports, and tracking placement trends.

Core Objectives

- **Visualize Alumni Outcomes:** Enable users to explore career paths of CS, DS, and AI graduates based on role, industry, graduate education, geography, and salary.

- **Track Salary & Placement Trends:** Display compensation and placement trends by academic track, graduation year, internship experience, and more.

- **Support Academic & Career Planning:** Empower students and advisors with real-world data to guide decisions around majors, internships, and job opportunities.

- **Showcase Departmental Success:** Provide Purdue with a compelling tool to highlight alumni impact to prospective students, families, and corporate partners.

- **Enable Efficient Data Management:** Allow administrators to upload, clean, moderate, and analyze alumni data with export capabilities and permission controls.

Application Structure

The application will be split into two main user experiences: one for students/ advisors/ visitors and one for administrators managing the data and alumni inputs.

For Purdue Faculty + Students:

- **Homepage**
    - Overview of the dashboard's purpose
    - Highlight stats: # of alumni tracked, industries, top companies
    - Quick entry points: "Explore Data," "Find Alumni"
- **Career Outcomes Examples Dashboard**
    - Salary Trends by Track & Role
        - View average, median, and range of salaries for CS, DS, and AI graduates.
            - Filterable, interactive dashboards based on:
                - Major
                - Track
                - Graduation year
                - Immediate Employment/ Post Graduate School
                - Residency
                - Company or role
                - Geographic location
    - Potential visualizations include-
        - Sankey diagrams (degree → role → company flow)
        - Bar/line charts (salary trends over time)
        - Heat maps (location-based placement data)
        - Pie charts (industry/grad school breakdowns)
    - Internship Outcome Explorer
        - Visualize how specific internships led to full-time offers, role types, or transitions into graduate programs.
        - Includes a "Where did they go after X company?" view to show long-term patterns from initial internship placements.
- **Alumni Directory**
    - Browseable profiles of alumni (opt-in only)
        - Filters: graduation year, company, role, mentorship availability
        - Profile fields: LinkedIn, employer, position, location, personal insights

For Admin (Faculty, CCO Staff, Development Officers):

- **Data Management Module**
  - Upload new data via CSV or API (from LinkedIn web-scraping or Purdue4Life)
  - Validate entries with auto-detection for duplicates, missing fields
  - Tag entries by track, employer, degree, etc.
- **Alumni Profile Moderation**
  - Approve, edit, or anonymize submitted alumni info
  - Set visibility for fields like salary, contact info
  - Link profiles to public-facing Alumni Explorer
- **Export & Reporting**
  - Generate PDF/CSV reports by filter selection (e.g., "2022 DS grads who interned at startups")
  - Share dashboards or snapshots with corporate partners

Proposed Tech Stack

- **Frontend**
  - React - has libraries for creating dynamic and interactive diagrams and charts
  - HTML/CSS/Javascript
- **Backend**
  - Node.js with Express.js
- **Database**
  - PostgreSQL or integrate with an existing database if maintained by Purdue
- Integrate with **Purdue University's Single Sign On** for authentication and login

Feature Suggestions

- Alumni Mentorship Tool
  - Filterable directory of alumni who have opted into mentorship roles (e.g., mock interviews, resume reviews, casual career chats).
  - Filters: track, graduation year, role, location, availability (ongoing vs. one-time).
  - Built-in message templates for students to send contact requests via email or LinkedIn.
  - Mentorship Request Tracker: students can save and track their outreach efforts.